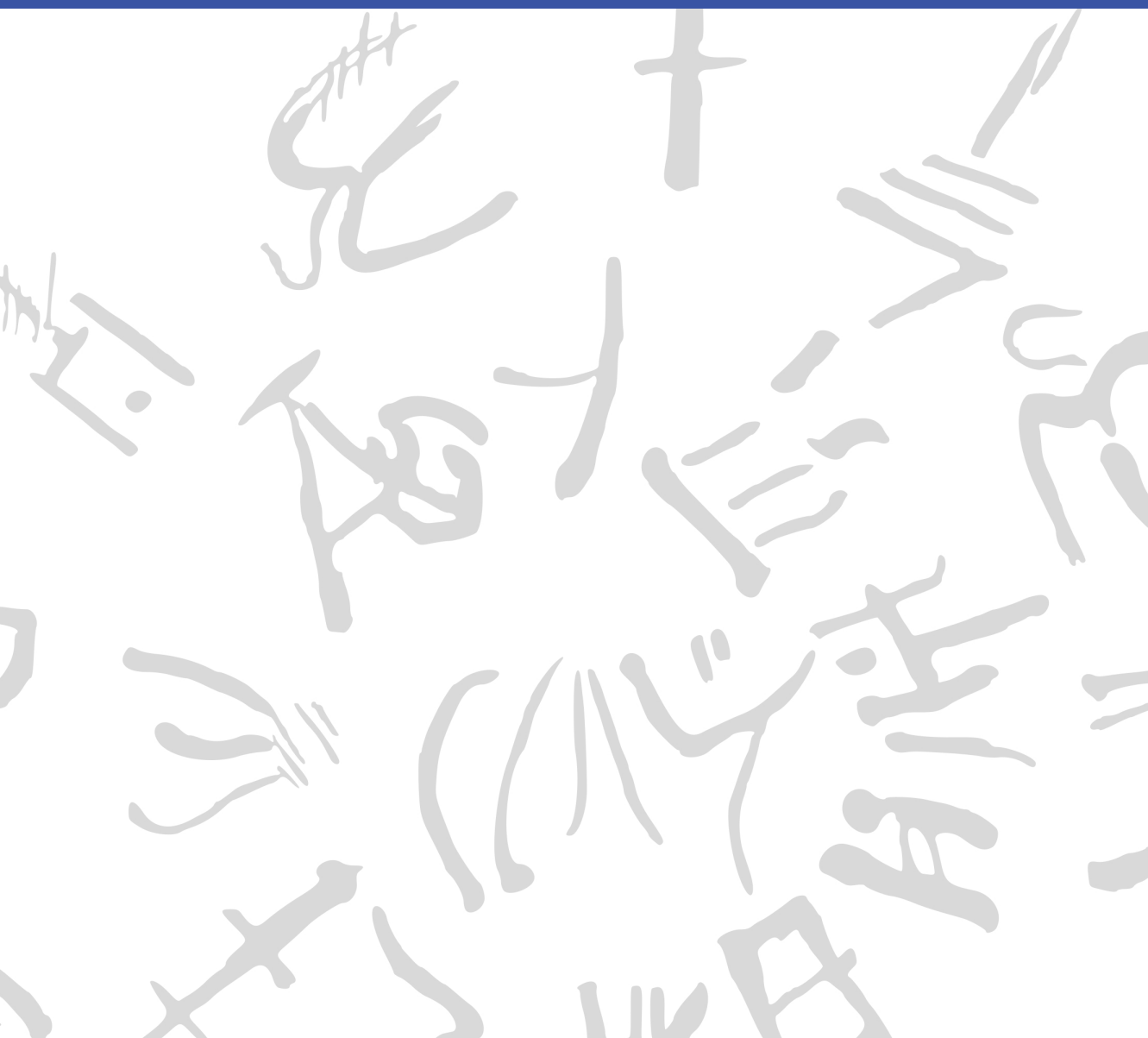


Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

Bachelor Thesis



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Preface

This Bachelor's thesis was prepared at the Department of Applied Mathematics and Computer Science at the Technical University of Denmark in fulfillment of the requirements for acquiring a Bachelor of Science in Engineering (BSc Eng) degree in General Engineering (Cyber Systems).

Kongens Lyngby, March 1, 2026

A handwritten signature in black ink, enclosed within a roughly drawn, rounded rectangular border. The signature appears to be 'F. W. L. Christoffersen' written in a stylized, cursive-like font.

Frederik W. L. Christoffersen

Abstract

This thesis contains a thorough description of the problem to be solved, including the context necessary to fathom the importance of solving it. It also outlines an initial miniature literature review that provides an overview of the most relevant work and sources to be further studied and cited. Finally, it covers the associated plan and timeline that will guide the project's execution, including a Gantt chart and key milestones.

Acknowledgements

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Generative AI tools were occasionally used as support for brainstorming, linguistic editing, code exploration, and research during the development of this project.

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Acronyms

SAM Segment Anything Model

Cefael Collections de l'École française d'Athènes en ligne

PDR Project Definition Report

IPR Intellectual Property Rights

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CHAPTER 1

Introduction

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, currently consisting of only around 1,500 documents - significantly smaller than that of its younger counterpart, which now spans more than 6,000 documents.¹ But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.² Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael) (see Godart and Olivier (1976–1985)), these are only available in print form.

This motivated the development of the palaeographical³ database *The Signs of Linear A* (SigLA), which seeks to "[develop] a systematic, exhaus-

¹SteeleAegeanWriting2023.

²Consiglio Nazionale delle Ricerche 2025.

³The term "palaeographical" refers to the study of ancient writing systems.

tive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palæographic analyses within the epigraphic corpus” (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known documents.^{4,5}

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.⁶ According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation tool can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA’s database.

1.1.2 A problem of small data

The small size of the Linear A corpus and the irregularities in quality of its documents is a huge problem for the application of deep learning approaches, which typically require large amounts of data to perform well. Consequently, this project will instead seek to make use of generalized machine learning and computer vision, in particular Meta’s state-of-the-art interactive segmentation tool, the Segment Anything Model (SAM) (Kirillov et al. 2023), which is pre-trained on a massive and diverse dataset of over 1 billion masks. SAM’s generalization and interactivity

⁴SigLA 2021a.

⁵SigLA 2021b.

⁶Salgarella and Castellan 2020.

(allowing user-placed markers that guide the model) might thus allow for semi-automatic annotation of Linear A inscriptions without the need for large amounts of training data, which is unfortunately not available in this context.⁷

1.2 Prior work / Literature review

This thesis places itself in the intersection of computer vision, digital humanities, and epigraphy (the study of inscriptions). Since it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. As a result, the other two fields will merely provide the context and motivation for the research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM, etc.), 2. the palæographical and digital humanities work that this project will seek to support (SigLA), and 3. the overall context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (**SteeleAegeanWriting2023**), its new addition (Consiglio Nazionale delle Ricerche 2025), and where it's accessed (Godart and Olivier 1976–1985). And for the epigraphical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (Salgarella and Castellan 2020), its about page (SigLA 2021a), and its document repository (SigLA 2021b). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

⁷By the "interactivity" of SAM, what is meant is NOT any built-in user interface in the form of a GUI, but rather the possibility of prompt-based placement of rectangles of focus, and points of additions and deletions that guide the model's prediction.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (Gonzalez and Woods 2018) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (Otsu 1979), and of course the OpenCV Python documentation (OpenCV Team 2024) for short implementation reference. Then, for SAM are: its paper (Kirillov et al. 2023), its lighter implementation MobileSAM (C. Zhang et al. 2023) - and the newest version of that one MobileSAMv2 (Zhao et al. 2023). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art (P. Zhang, Li, and Sun 2024), and a lighter EU-funded historical document segmentation tool *dhSegment* (Oliveira, Seguin, and Kaplan 2018) also using deep learning.

1.3 Problem statement

As previously stated, the aim of this project is to investigate whether parts of the process of creating annotated drawings for SigLA can be automated. More specifically, whether some kind of semi-automatic segmentation tool using generalized computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database. This is a problem that can be divided into the following three sub-tasks to be solved: 1. the pre-processing of original document images to make them suitable for SAM, 2. the incorporation of SAM into a segmentation tool potentially with the possibility of optional user interaction whose exports can be integrated into SigLA's workflow, and 3. the quantitative evaluation of the tool's performance compared to baselines against metrics of segmentation quality.

1.4 Empirical considerations

For the evaluation of the solution, a dataset will be compiled together with Ester Salgarella of at least 25 images of original documents from the corpus, selected and cropped from Cefael. These images are subject to Intellectual Property Rights (IPR) issues, and can thus not be displayed publicly if not through Cefael. However, they can still be used for the development and evaluation of the tool (and can be bought if necessary for show-casing within the thesis).

As ground truth, their corresponding digitized layered drawings, stored as Krita (.kra) files on SigLA (provided by Ester Salgarella) will be used and aligned for comparison with tool outputs. The ground truth data might then have to be cleared of any marked “deleted signs” or other annotation elements that our model is not intended to reproduce.

The solution will then be built as a pipeline, consisting of pre-processing, segmentation, post-processing, and export (e.g. as Krita files). At last, the primary deliverable of this project will be delivered to the external partner and co-supervisor Ester Salgarella in the form of a proof-of-concept semi-automatic annotation tool. All supporting components, including evaluation results, and analysis will be documented and presented in the written thesis.

1.5 Impact – innovation and application

The work of this thesis will present itself as a first proof-of-concept for the use of modern computer vision to accelerate the expansion of the SigLA database, supporting the further digitization of the Linear A corpus. Its main contribution will be a semi-automatic annotation tool, which might prove useful to epigraphers for generating segmentations of original document images, as they can then be completed and refined more easily. Hopefully, it will inspire further research and development within the field of digital epigraphy, particularly in the application of generalized and interactive computer vision approaches such as SAM on Linear A,

and perhaps even in the broader context of use on small corpus ancient writing systems where the lack of big data is ill fit for approaches using deep learning.

CHAPTER 2

Methodology

2.1 Common methodology

- Data assumptions (brief reference to Data chapter)
- Preprocessing pipeline
- General evaluation metrics (IoU, Dice, etc.)
- Experimental conventions

2.2 Segmentation quality

2.2.1 Baseline model configuration

2.2.2 State-of-the-art model configuration

2.2.3 Evaluation protocol and performance metrics

2.3 Interactive semi-automation

2.3.1 Interactive tool architecture

2.3.2 Experimental design for automation evaluation

2.3.3 Definition of automation metrics

2.4 Workflow efficiency

2.4.1 Definition of workflow efficiency metrics

2.4.2 Experimental design of the user study

2.4.3 Analysis of results

CHAPTER 3

Data

- 3.1 Data sources
- 3.2 Dataset construction
- 3.3 Ground truth definition
- 3.4 Dataset characteristics
- 3.5 Difficulty stratification
- 3.6 Dataset limitations and biases

CHAPTER 4

Results

- 4.1 Segmentation performance
- 4.2 Automation evaluation
- 4.3 Workflow efficiency evaluation

CHAPTER 5

Discussion

5.1 Segmentation ability

5.2 Automation and interactivity

5.3 Workflow impact

CHAPTER 6

Conclusion

6.1 Summary of findings per research question

6.1.1 RQ1: Segmentation performance

6.1.2 RQ2: Degree of automation

6.1.3 RQ3: Workflow efficiency

6.2 Overall contribution

6.3 Final remarks

CHAPTER 7

Future Work

- 7.1 Methodological extensions
- 7.2 Dataset expansion and diversification
- 7.3 Further automation opportunities

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APPENDIX A

An Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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